

PBSAI Control Mapping to the NIST AI Risk Management Framework (AI RMF)

A Multi-Agent Governance Architecture for Secure Enterprise AI Estates

Abstract

The Practitioner's Blueprint for Secure AI (PBSAI) Governance Ecosystem is a multi-agent governance and operational security architecture designed to secure enterprise Artificial Intelligence (AI) estates through deterministic governance mediation, evidence-centric accountability, bounded operational autonomy, and cross-domain orchestration.

This document provides a comprehensive mapping between PBSAI governance controls and the National Institute of Standards and Technology (NIST) Artificial Intelligence Risk Management Framework (AI RMF). The mapping demonstrates how PBSAI operationalizes AI RMF governance principles through runtime governance enforcement, structured policy mediation, provenance-aware execution, explainable operational workflows, human oversight controls, and continuous assurance mechanisms.

PBSAI organizes enterprise AI governance across twelve operational domains spanning:

- governance and compliance,
- identity and access management,
- telemetry and threat monitoring,
- protective enforcement,
- data governance,
- incident response,
- resilience,
- secure architecture,
- physical security,
- supply chain security,
- and AI assurance orchestration.

Unlike governance models that remain primarily organizational or policy-centric, PBSAI provides an operational realization architecture capable of mediating AI-enabled actions across enterprise systems, orchestration environments, and cybersecurity operations.

The mapping illustrates how PBSAI aligns with:

- the AI RMF GOVERN function through governance orchestration and accountability,
 - the MAP function through contextual awareness and dependency mapping,
 - the MEASURE function through telemetry analytics, evidence generation, and validation,
 - and the MANAGE function through bounded remediation, resilience orchestration, and adaptive governance.
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1. Introduction

Enterprise AI ecosystems increasingly integrate:

- large language models (LLMs),
- retrieval-augmented generation (RAG),
- autonomous agents,
- orchestration platforms,
- operational analytics,
- infrastructure automation,
- and AI-enabled cybersecurity systems.

These systems operate across:

- cloud platforms,
- enterprise infrastructure,
- identity systems,
- operational technology,
- data pipelines,
- security operations centers,
- and distributed orchestration environments.

Traditional governance approaches frequently emphasize:

- organizational oversight,
- policy frameworks,

- and lifecycle documentation.

However, operational AI systems increasingly require:

- runtime governance mediation,
- deterministic authorization,
- bounded execution,
- explainable operational behavior,
- evidence-centric accountability,
- and continuous assurance validation.

PBSAI addresses these requirements through:

- structured governance controls,
- machine-readable policy mediation,
- provenance-aware operational validation,
- deterministic execution governance,
- evidence-centric telemetry correlation,
- and bounded AI-enabled orchestration.

The architecture assumes that:

- AI reasoning may remain probabilistic, while:
- operational execution authority must remain deterministic, governable, explainable, and auditable.

Accordingly, PBSAI separates:

- AI-generated recommendations, from:
- operational execution authority.

AI outputs are therefore treated as proposed operational actions requiring:

- policy validation,
- authorization mediation,
- provenance verification,

- trust-boundary enforcement,
- and governance approval prior to execution.

2. PBSAI Governance Architecture Overview

PBSAI organizes governance responsibilities into twelve operational domains.

2.1 PBSAI Operational Domains

Domain	Description
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Domain A	Governance, Risk, and Compliance
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Domain B	Asset, Configuration, and Change Governance
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Domain C	Identity, Credential, and Access Management
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Domain D	Threat Intelligence, Monitoring, and Situational Awareness
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Domain E	Protective Enforcement and Response
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Domain F	Data Security and Privacy
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Domain G	Incident Response and Forensics
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Domain H	Resilience, Continuity, and Recovery
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Domain I	Security Architecture and Systems Engineering
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Domain J	Physical and Environmental Security
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Domain K	Supply Chain and Lifecycle Security
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Domain L	Security Program, Analytics, Orchestration, and AI Assurance
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2.2 Global PBSAI Governance Controls

PBSAI also defines global governance controls that operate across all domains.

Global Governance Control	Purpose
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PBSAI-GEN-MCP	Context envelope governance
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Global Governance Control Purpose

PBSAI-GEN-OC	Structured Output Contract enforcement
PBSAI-GEN-EVID	Evidence generation
PBSAI-GEN-PROV	Provenance and attribution
PBSAI-GEN-HITL	Human authorization and oversight
PBSAI-GEN-SAFE	Safe execution and rollback semantics
PBSAI-GEN-FEED	Continuous feedback and improvement

2.3 Core Governance Principles

PBSAI operationalizes:

- deterministic governance mediation,
 - evidence-centric accountability,
 - explainable operational workflows,
 - provenance-aware authorization,
 - bounded autonomous execution,
 - cross-domain telemetry correlation,
 - human oversight enforcement,
 - continuous assurance,
 - and resilient operational governance.
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3. PBSAI and NIST AI RMF Trustworthiness Characteristics

PBSAI directly supports multiple trustworthiness characteristics identified by the NIST AI RMF.

3.1 Trustworthiness Characteristics Alignment

AI RMF Trustworthiness Characteristic PBSAI Alignment

Accountability	Very Strong
Explainability and Interpretability	Strong
Privacy Enhancement	Strong
Reliability	Strong
Safety	Very Strong
Security and Resilience	Very Strong
Transparency	Strong
Fairness	Moderate
Governability	Very Strong
Traceability	Very Strong
Human Oversight	Very Strong

3.2 Explainability and Operational Transparency

PBSAI supports explainability through:

- structured Output Contracts,
- confidence scoring,
- rationale preservation,
- telemetry correlation,
- operational evidence generation,
- and deterministic authorization mediation.

Operational actions remain attributable through:

- provenance records,
- governance evidence,
- authorization chains,
- and structured telemetry artifacts.

3.3 Accountability and Human Oversight

PBSAI enforces:

- human authorization controls,
- escalation workflows,
- multi-party approval requirements,
- bounded automation semantics,
- and deterministic governance validation.

High-impact operational actions require:

- policy validation,
- governance mediation,
- and explicit authorization prior to execution.

3.4 Security and Resilience

PBSAI operationalizes:

- bounded remediation,
- rollback semantics,
- resilience orchestration,
- threat-adaptive monitoring,
- segmentation governance,
- trust-boundary enforcement,
- and deterministic recovery coordination.

These controls support:

- secure operational continuity,
 - explainable incident response,
 - and resilient AI-enabled enterprise operations.
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4. PBSAI Alignment to AI RMF Functions

The NIST AI RMF defines four primary functions:

1. GOVERN
2. MAP
3. MEASURE
4. MANAGE

PBSAI aligns operationally with these functions through its domain-partitioned governance architecture.

4.1 PBSAI Domain Alignment to AI RMF Functions

AI RMF Function	Primary PBSAI Domains	Primary Responsibilities
GOVERN	A, I, L	Governance policy, architecture governance, AI oversight, operational governance
MAP	B, C, F, K	Asset awareness, contextual authorization, provenance, data governance
MEASURE	D, L	Monitoring, telemetry analytics, validation, AI assurance
MANAGE	E, G, H, L	Response, remediation, resilience, orchestration, recovery

4.2 PBSAI Operational Realization of AI Governance

PBSAI transforms:

- governance objectives, into:
- runtime operational governance controls.

This includes:

- contextual authorization,
- structured policy mediation,

- bounded execution,
- explainable orchestration,
- evidence generation,
- provenance validation,
- and continuous assurance monitoring.

5. PBSAI Control Mapping to NIST AI RMF — GOVERN Function

The GOVERN function addresses:

- organizational governance,
- accountability,
- oversight,
- policy management,
- operational governance,
- risk management,
- transparency,
- and trustworthy AI governance structures.

PBSAI aligns strongly with the GOVERN function through:

- Domain A — Governance, Risk, and Compliance,
- Domain I — Security Architecture and Systems Engineering,
- Domain L — Security Program and AI Assurance,
- and the PBSAI global governance control layer.

PBSAI operationalizes governance not only through policy definition, but also through:

- runtime authorization mediation,
- deterministic execution validation,
- structured evidence generation,
- provenance-aware governance,
- and bounded operational autonomy.

Correct.

The proper structure should be:

5. GOVERN Function Mapping

5.1 GOVERN — Policies, Processes, and Oversight

Then the normalized AGCP-style control mappings underneath.

For example:

5. GOVERN Function Mapping

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- Domain A — Governance, Risk, and Compliance,
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 - and the PBSAI global governance control layer.
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5.1 GOVERN — Policies, Processes, and Oversight

PBSAI-A-GOV-001 — Governance Policy Management

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 1.1	Accountability	equivalent	Establishes enterprise AI

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance policies and operational governance requirements
GOVERN	GOVERN 1.2	Governability	strongly-supports	Defines governance authority and policy enforcement structures
GOVERN	GOVERN 1.5	Transparency	supports	Enables traceable governance documentation and policy accountability

PBSAI-A-GOV-002 — Governance Roles and Responsibilities

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 1.2	Accountability	equivalent	Defines accountable governance and oversight responsibilities
GOVERN	GOVERN 1.6	Human Oversight	strongly-supports	Establishes operational review and escalation authority
GOVERN	GOVERN 2.1	Traceability	supports	Maintains attributable

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance responsibilities

PBSAI-A-RISK-001 — AI Risk Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 1.4	Governability	equivalent	Integrates AI operational risk into enterprise governance processes
MAP	MAP 4.1	Transparency	strongly-supports	Establishes contextual operational risk awareness
MANAGE	MANAGE 1.1	Reliability	supports	Guides mitigation prioritization and operational treatment

PBSAI-GEN-HITL-001 — Human Approval Requirement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 1.6	Human Oversight	equivalent	Requires human authorization for high-impact operational actions
MANAGE	MANAGE 5.1	Safety	strongly-supports	Prevents unsafe autonomous

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				operational execution
MANAGE	MANAGE 5.2	Accountability	supports	Preserves attributable authorization workflows
MAP	MAP 2.4	Governability	supports	Associates stakeholders and authorization responsibilities with execution context

5.2 GOVERN — Accountability, Transparency, and Explainability

PBSAI-GEN-EVID-001 — Evidence Generation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.1	Accountability	equivalent	Produces attributable governance evidence for operational actions
MEASURE	MEASURE 3.1	Traceability	equivalent	Generates structured operational telemetry and audit evidence
MEASURE	MEASURE 3.3	Transparency	strongly- supports	Enables explainable operational

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				reconstruction and review
MANAGE	MANAGE 4.1	Reliability	supports	Supports adaptive governance improvement through operational evidence

PBSAI-GEN-PROV-001 — Provenance and Attribution

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.2	Accountability	equivalent	Maintains attributable provenance for operational actions
MEASURE	MEASURE 3.2	Traceability	equivalent	Preserves provenance-linked operational history
MAP	MAP 4.3	Transparency	strongly-supports	Enables dependency and provenance visibility
MANAGE	MANAGE 2.4	Reliability	supports	Supports trustworthy remediation and forensic reconstruction

PBSAI-GEN-OC-001 — Structured Output Contracts

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.3	Explainability	equivalent	Standardizes machine-readable operational outputs
MEASURE	MEASURE 2.3	Transparency	strongly-supports	Preserves explainable operational semantics
MEASURE	MEASURE 3.1	Traceability	supports	Produces structured evidence-compatible output artifacts
MANAGE	MANAGE 5.3	Safety	supports	Constrains operational execution through deterministic schema validation

PBSAI-GEN-MCP-001 — Context Envelope Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 2.3	Traceability	equivalent	Preserves contextual metadata and governance bindings

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.2	Accountability	strongly-supports	Associates operational actions with contextual governance state
MEASURE	MEASURE 2.2	Transparency	supports	Enables explainable contextual evaluation and operational interpretation
MANAGE	MANAGE 5.4	Governability	supports	Preserves governance constraints during operational execution

PBSAI-L-AI-003 — AI Confidence and Explainability Enforcement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.3	Explainability	equivalent	Preserves confidence scoring and rationale metadata
MEASURE	MEASURE 2.3	Transparency	equivalent	Enables explainable operational interpretation
MEASURE	MEASURE 2.4	Reliability	strongly-supports	Supports detection of

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				anomalous AI behavior
MANAGE	MANAGE 4.2	Governability	supports	Enables adaptive governance based on explainability evaluation

PBSAI-L-ANL-002 — Cross-Domain Telemetry Correlation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.4	Transparency	supports	Supports operational transparency through telemetry aggregation
MEASURE	MEASURE 4.1	Traceability	equivalent	Correlates telemetry and operational evidence across governance domains
MEASURE	MEASURE 4.2	Reliability	strongly-supports	Enables cross-domain operational analytics and anomaly detection
MANAGE	MANAGE 4.1	Governability	supports	Enables adaptive governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				informed by operational telemetry

PBSAI-G-EVID-003 — Response Attribution and Accountability

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.1	Accountability	strongly-supports	Maintains attributable operational response history
MEASURE	MEASURE 3.2	Traceability	supports	Preserves forensic linkage between actions and actors
MANAGE	MANAGE 2.4	Reliability	supports	Supports trustworthy response reconstruction and investigation

PBSAI-F-EVID-002 — Data Lineage and Handling Traceability

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.2	Accountability	supports	Maintains traceability for governed data operations
MAP	MAP 3.3	Transparency	equivalent	Preserves contextual

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				visibility into data lineage and handling
MEASURE	MEASURE 3.2	Traceability	strongly-supports	Enables reconstruction of governed data operations
MANAGE	MANAGE 4.1	Reliability	supports	Supports trustworthy remediation and governance adaptation

PBSAI-L-AI-006 — AI Assurance Evidence and Traceability

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.1	Accountability	equivalent	Produces attributable evidence for AI operational behavior
MEASURE	MEASURE 2.5	Traceability	equivalent	Preserves explainable AI governance evidence
MEASURE	MEASURE 3.3	Transparency	strongly-supports	Enables explainable AI operational reconstruction
MANAGE	MANAGE 4.2	Governability	supports	Supports adaptive AI governance and

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				assurance workflows

5.3 GOVERN — Governance Enforcement and Operational Constraints

PBSAI-GEN-SAFE-001 — Safe Execution Constraint

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.1	Safety	equivalent	Enforces bounded operational execution semantics
MANAGE	MANAGE 1.3	Reliability	strongly- supports	Prevents unsafe or uncontrolled operational behavior
MANAGE	MANAGE 5.3	Governability	supports	Constrains orchestration workflows within approved operational limits
MEASURE	MEASURE 1.2	Security and Resilience	supports	Enables validation of operational safety conditions

PBSAI-GEN-SAFE-002 — Rollback Requirement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.2	Safety	strongly-supports	Enables recovery from unsafe operational outcomes
MANAGE	MANAGE 3.4	Reliability	equivalent	Supports controlled rollback and operational restoration
MANAGE	MANAGE 5.3	Security and Resilience	strongly-supports	Preserves resilient operational continuity during failure conditions
MEASURE	MEASURE 1.5	Reliability	supports	Enables validation of rollback readiness and restoration capability

PBSAI-GEN-SAFE-003 — Kill Switch / Emergency Stop

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.4	Safety	equivalent	Enables rapid interruption of unsafe operational behavior
GOVERN	GOVERN 3.1	Governability	strongly-supports	Maintains deterministic

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.2	Security and Resilience	supports	override authority Supports containment during compromise or operational instability
MEASURE	MEASURE 1.1	Reliability	supports	Enables validation of emergency interruption capability

PBSAI-E-ENF-001 — Policy Enforcement Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.1	Governability	equivalent	Enforces policy-bound operational execution
MANAGE	MANAGE 1.1	Safety	strongly-supports	Applies protective enforcement constraints to operational workflows
MANAGE	MANAGE 5.2	Accountability	supports	Preserves attributable enforcement authorization records

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.1	Reliability	supports	Enables validation of enforcement effectiveness

PBSAI-E-ENF-002 — Response Authorization Validation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.1	Accountability	equivalent	Validates authorization before remediation or response execution
MANAGE	MANAGE 5.1	Human Oversight	strongly-supports	Requires governed authorization prior to operational action
MAP	MAP 2.2	Governability	supports	Associates operational response authority with contextual authorization
MEASURE	MEASURE 3.2	Traceability	supports	Preserves attributable response authorization history

PBSAI-E-ENF-003 — Response Scope Constraint

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.1	Safety	equivalent	Constrains operational remediation and enforcement scope
MANAGE	MANAGE 1.3	Governability	strongly-supports	Prevents operational actions from exceeding approved boundaries
MAP	MAP 1.3	Transparency	supports	Preserves trust-boundary and segmentation awareness
MEASURE	MEASURE 4.2	Reliability	supports	Enables validation of bounded remediation behavior

PBSAI-G-CONT-001 — Incident Containment Execution

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.2	Security and Resilience	equivalent	Executes bounded operational containment actions
GOVERN	GOVERN 3.1	Safety	strongly-supports	Maintains governed containment

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				execution authority
MEASURE	MEASURE 1.1	Reliability	supports	Enables validation of containment effectiveness
MAP	MAP 4.2	Transparency	supports	Preserves contextual awareness during containment workflows

PBSAI-G-CONT-003 — Containment Scope Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.1	Governability	equivalent	Constrains containment authority within approved operational limits
MANAGE	MANAGE 5.2	Safety	strongly-supports	Prevents unsafe or excessive containment actions
MAP	MAP 1.3	Transparency	supports	Maintains trust-boundary visibility during containment

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.1	Reliability	supports	Enables validation of governed containment behavior

PBSAI-I-DEP-003 — Deployment Scope Constraints

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.2	Governability	equivalent	Restricts deployment authority and operational scope
MANAGE	MANAGE 5.2	Safety	strongly-supports	Prevents unauthorized or unsafe deployment execution
MAP	MAP 1.2	Transparency	supports	Preserves deployment-context visibility
MEASURE	MEASURE 3.4	Traceability	supports	Enables attributable deployment governance records

PBSAI-I-DEP-004 — Rollback and Safe Release Management

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.2	Safety	equivalent	Governs rollback semantics for deployment operations
MANAGE	MANAGE 3.4	Reliability	strongly-supports	Enables controlled recovery from unsafe deployments
MANAGE	MANAGE 5.3	Security and Resilience	supports	Preserves operational continuity during failed release conditions
MEASURE	MEASURE 1.5	Reliability	supports	Validates rollback readiness and deployment recovery capability

PBSAI-L-ORCH-004 — Autonomous Workflow Constraint Enforcement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.1	Governability	equivalent	Constrains orchestration workflows within policy-defined boundaries
MANAGE	MANAGE 5.3	Safety	strongly-supports	Prevents unsafe autonomous

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				operational execution
MEASURE	MEASURE 4.1	Reliability	supports	Enables validation of orchestration governance behavior
MAP	MAP 2.3	Transparency	supports	Preserves contextual governance constraints during orchestration

PBSAI-H-REC-006 — Recovery Rollback and Safe Restoration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.2	Safety	strongly-supports	Maintains safe operational restoration behavior
MANAGE	MANAGE 3.3	Reliability	equivalent	Supports controlled recovery and restoration workflows
MANAGE	MANAGE 3.4	Security and Resilience	strongly-supports	Preserves resilient continuity during recovery operations

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.5	Reliability	supports	Enables validation of restoration and rollback readiness

5.4 GOVERN — Organizational Risk Management and Continuous Governance

PBSAI-A-RISK-002 — Enterprise Risk Assessment

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.1	Governability	equivalent	Defines enterprise AI risk evaluation and governance assessment processes
MAP	MAP 4.1	Transparency	strongly-supports	Establishes contextual operational risk awareness
MEASURE	MEASURE 4.2	Reliability	supports	Enables continuous evaluation of operational exposure and systemic risk
MANAGE	MANAGE 1.1	Security and Resilience	supports	Guides mitigation prioritization and operational treatment planning

PBSAI-A-RISK-003 — Risk Prioritization and Treatment

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.2	Governability	equivalent	Prioritizes AI operational risk treatment and mitigation activities
MANAGE	MANAGE 1.1	Reliability	strongly-supports	Aligns operational mitigation with enterprise risk posture
MEASURE	MEASURE 4.2	Transparency	supports	Preserves explainable risk-treatment rationale
MAP	MAP 4.1	Accountability	supports	Associates mitigation decisions with contextual risk assessments

PBSAI-L-PROG-002 — Program Risk Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.1	Accountability	equivalent	Integrates operational risk into enterprise governance programs
GOVERN	GOVERN 1.5	Governability	strongly-supports	Aligns program governance with

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				enterprise AI risk strategy
MANAGE	MANAGE 4.3	Reliability	supports	Supports adaptive governance improvement using operational risk findings
MEASURE	MEASURE 4.2	Transparency	supports	Enables traceable program-level risk analytics

PBSAI-L-ASSUR-001 — Continuous Assurance Evaluation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.3	Accountability	equivalent	Continuously evaluates governance and operational assurance posture
MEASURE	MEASURE 1.2	Reliability	strongly-supports	Validates operational trustworthiness and governance effectiveness
MEASURE	MEASURE 4.4	Security and Resilience	equivalent	Correlates assurance evidence across operational domains

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.2	Governability	supports	Enables adaptive governance response to assurance degradation

PBSAI-L-ASSUR-003 — Assurance Drift Detection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.3	Reliability	strongly-supports	Detects degradation in governance and operational assurance posture
MEASURE	MEASURE 2.4	Security and Resilience	equivalent	Continuously monitors assurance integrity and operational drift
MEASURE	MEASURE 4.4	Transparency	supports	Enables explainable detection of assurance degradation
MANAGE	MANAGE 4.2	Governability	supports	Supports adaptive remediation of assurance deficiencies

PBSAI-L-FEED-001 — Operational Feedback Collection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.4	Accountability	supports	Collects operational findings and governance outcomes
MEASURE	MEASURE 5.1	Transparency	equivalent	Preserves operational feedback for governance evaluation
MANAGE	MANAGE 4.1	Reliability	strongly-supports	Supports adaptive operational improvement workflows
MANAGE	MANAGE 4.3	Governability	supports	Enables continuously adaptive governance processes

PBSAI-L-FEED-003 — Control Improvement Recommendation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.4	Governability	equivalent	Recommends governance improvements using operational evidence
MEASURE	MEASURE 5.2	Transparency	strongly-supports	Preserves explainable governance-

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				improvement rationale
MANAGE	MANAGE 4.2	Reliability	supports	Enables adaptive mitigation and operational optimization
MEASURE	MEASURE 4.2	Accountability	supports	Associates recommendations with operational evidence and findings

PBSAI-L-FEED-004 — Governance Adaptation and Optimization

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.4	Governability	equivalent	Continuously adapts governance workflows and operational controls
MANAGE	MANAGE 4.3	Reliability	strongly-supports	Enables resilient and adaptive governance evolution
MEASURE	MEASURE 5.3	Transparency	supports	Preserves explainable governance adaptation decisions

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.4	Security and Resilience	supports	Supports continuous resilience-oriented governance optimization

PBSAI-D-THREAT-003 — Threat-Adaptive Monitoring

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.3	Security and Resilience	supports	Adapts governance monitoring to evolving operational threats
MAP	MAP 4.2	Transparency	strongly-supports	Integrates evolving threat context into operational awareness
MEASURE	MEASURE 1.4	Reliability	equivalent	Dynamically adjusts monitoring based on operational threat conditions
MANAGE	MANAGE 4.1	Governability	supports	Enables adaptive governance response to

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				changing threats

PBSAI-K-SCRM-001 — Supply Chain Risk Assessment

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.2	Accountability	supports	Integrates supplier and dependency risk into governance evaluation
MAP	MAP 4.3	Transparency	equivalent	Evaluates supplier and dependency exposure relationships
MEASURE	MEASURE 4.2	Reliability	supports	Assesses systemic operational exposure from external dependencies
MANAGE	MANAGE 1.1	Security and Resilience	strongly-supports	Guides mitigation and treatment of supply-chain operational risk

PBSAI-H-TEST-005 — Improvement and Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.4	Reliability	supports	Uses resilience-testing outcomes to improve governance posture
MEASURE	MEASURE 5.2	Accountability	strongly-supports	Integrates testing evidence into governance evaluation workflows
MANAGE	MANAGE 3.5	Security and Resilience	equivalent	Continuously improves resilience and recovery readiness
MANAGE	MANAGE 4.3	Governability	supports	Enables adaptive continuity and resilience governance improvement

5.5 GOVERN — Cross-Domain Governance Integration

PBSAI operationalizes governance as:

- deterministic authorization mediation,
- evidence-centric accountability,
- provenance-aware operational validation,
- explainable orchestration,
- bounded autonomous execution,
- and continuously adaptive governance assurance.

The GOVERN alignment demonstrates that PBSAI continuously governs:

- policy enforcement,
- operational accountability,
- runtime authorization,
- AI oversight,
- resilience governance,
- orchestration constraints,
- and enterprise risk adaptation.

Unlike governance models limited to procedural oversight or documentation, PBSAI introduces:

- runtime governance enforcement,
- machine-readable policy mediation,
- explainable operational constraints,
- structured evidence generation,
- trust-boundary enforcement,
- and adaptive but bounded governance orchestration.

This enables:

- trustworthy AI-enabled enterprise operations,
- explainable operational governance,
- resilient authorization workflows,
- continuous assurance validation,
- and evidence-centric enterprise AI governance.

6. MAP Function Mapping

The MAP function addresses:

- contextual awareness,
- operational environment understanding,
- system and dependency mapping,
- stakeholder and authorization relationships,

- data governance context,
- supply-chain visibility,
- and operational risk framing.

PBSAI aligns strongly with MAP through:

- Domain B — Asset, Configuration, and Change Governance,
- Domain C — Identity, Credential, and Access Management,
- Domain F — Data Security and Privacy,
- Domain I — Security Architecture and Systems Engineering,
- Domain K — Supply Chain and Lifecycle Security,
- and the PBSAI context-governance control layer.

PBSAI operationalizes contextual governance through:

- structured context envelopes,
- provenance-aware metadata,
- identity-linked authorization,
- dependency inventories,
- trust-boundary modeling,
- and telemetry-aware operational correlation.

6.1 MAP — Asset, System, and Configuration Awareness

PBSAI-B-ASSET-001 — Asset Inventory Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 1.1	Transparency	equivalent	Maintains authoritative inventory of AI-enabled systems and operational assets

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.1	Accountability	strongly-supports	Associates operational governance responsibilities with managed assets
MEASURE	MEASURE 4.1	Traceability	supports	Enables telemetry and evidence correlation to governed assets
MANAGE	MANAGE 1.1	Reliability	supports	Supports risk treatment and operational mitigation planning

PBSAI-B-CONFIG-001 — Configuration Baseline Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 1.2	Transparency	equivalent	Tracks operational configuration state and deployment context
GOVERN	GOVERN 3.1	Governability	strongly-supports	Establishes controlled operational configuration governance
MEASURE	MEASURE 1.1	Reliability	supports	Enables monitoring of

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.3	Security and Resilience	supports	configuration drift and operational deviation Supports bounded operational remediation and restoration

PBSAI-B-CHANGE-001 — Change Governance and Tracking

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 1.2	Traceability	equivalent	Preserves contextual understanding of operational modifications
GOVERN	GOVERN 2.1	Accountability	strongly-supports	Maintains attributable operational change history
MEASURE	MEASURE 3.1	Transparency	supports	Produces explainable governance evidence for changes
MANAGE	MANAGE 4.1	Reliability	supports	Supports adaptive governance and remediation workflows

PBSAI-I-ARCH-003 — Trust Boundary Definition

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 1.3	Transparency	equivalent	Defines trust boundaries and operational segmentation context
GOVERN	GOVERN 3.1	Governability	strongly-supports	Enforces operational governance boundaries
MEASURE	MEASURE 4.1	Traceability	supports	Enables correlation of telemetry across segmented domains
MANAGE	MANAGE 1.1	Security and Resilience	supports	Limits operational exposure and propagation risk

PBSAI-I-SEG-001 — Security Segmentation Design

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 1.3	Transparency	equivalent	Maps segmented operational domains and interaction boundaries
GOVERN	GOVERN 3.1	Safety	strongly-supports	Enforces bounded operational

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				communication pathways
MEASURE	MEASURE 4.1	Reliability	supports	Enables monitoring and validation of segmentation effectiveness
MANAGE	MANAGE 2.2	Security and Resilience	supports	Supports containment and operational isolation workflows

PBSAI-K-SBOM-001 — SBOM Generation and Maintenance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 1.4	Transparency	equivalent	Maintains inventory and dependency awareness for software ecosystems
GOVERN	GOVERN 2.2	Accountability	strongly-supports	Preserves attributable dependency and component lineage
MEASURE	MEASURE 3.2	Traceability	supports	Enables explainable dependency and provenance validation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.1	Reliability	supports	Supports mitigation of dependency and supply-chain exposure

PBSAI-K-PROV-001 — Component Provenance Validation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 1.4	Traceability	equivalent	Tracks provenance and origin relationships for components and dependencies
GOVERN	GOVERN 2.2	Accountability	strongly-supports	Preserves attributable supply-chain lineage
MEASURE	MEASURE 3.2	Transparency	supports	Enables explainable provenance validation workflows
MANAGE	MANAGE 1.1	Security and Resilience	supports	Reduces operational risk from untrusted components

6.2 MAP — Identity, Authorization, and Stakeholder Context

PBSAI-C-ID-001 — Identity Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 2.1	Accountability	equivalent	Maintains identity awareness and authorization context
GOVERN	GOVERN 1.2	Governability	strongly-supports	Defines operational identity-governance responsibilities
MEASURE	MEASURE 3.2	Traceability	supports	Preserves attributable identity-linked operational evidence
MANAGE	MANAGE 5.1	Human Oversight	supports	Associates operational authority with governed identities

PBSAI-C-AUTH-001 — Authorization Policy Enforcement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 2.2	Governability	equivalent	Governs contextual authorization decisions
GOVERN	GOVERN 3.1	Accountability	strongly-supports	Enforces policy-bound authorization workflows

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.2	Safety	supports	Prevents unauthorized operational execution
MEASURE	MEASURE 3.2	Traceability	supports	Produces attributable authorization evidence

PBSAI-C-ROLE-001 — Role and Privilege Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 2.2	Accountability	equivalent	Maps operational privileges and governance roles
GOVERN	GOVERN 1.2	Governability	strongly-supports	Establishes controlled privilege-governance structures
MEASURE	MEASURE 3.1	Traceability	supports	Preserves attributable privilege assignment history
MANAGE	MANAGE 5.2	Safety	supports	Restricts operational authority within approved boundaries

PBSAI-C-SESS-001 — Session and Context Continuity Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 2.2	Traceability	equivalent	Maintains contextual continuity across operational workflows
GOVERN	GOVERN 2.2	Accountability	strongly-supports	Associates operational actions with continuous session context
MEASURE	MEASURE 2.2	Transparency	supports	Enables explainable contextual evaluation of operations
MANAGE	MANAGE 5.4	Governability	supports	Preserves governance constraints across long-running workflows

PBSAI-GEN-MCP-001 — Context Envelope Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 2.3	Traceability	equivalent	Preserves contextual metadata and governance bindings

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.2	Accountability	strongly-supports	Associates operational actions with contextual governance state
MEASURE	MEASURE 2.2	Transparency	supports	Enables explainable contextual evaluation and operational interpretation
MANAGE	MANAGE 5.4	Governability	supports	Preserves governance constraints during operational execution

6.3 MAP — Data, Privacy, and Information Governance Context

PBSAI-F-CLS-001 — Data Discovery

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.1	Transparency	equivalent	Identifies and maps governed data assets across enterprise environments
GOVERN	GOVERN 2.1	Accountability	strongly-supports	Associates governance responsibilities

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				with discovered data assets
MEASURE	MEASURE 3.2	Traceability	supports	Enables attributable tracking of governed data sources
MANAGE	MANAGE 1.1	Reliability	supports	Supports operational risk treatment and data-governance enforcement

PBSAI-F-CLS-002 — Data Classification

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.1	Transparency	equivalent	Establishes contextual data classification awareness
GOVERN	GOVERN 1.1	Governability	strongly-supports	Defines policy-driven data-governance controls
MEASURE	MEASURE 2.2	Traceability	supports	Enables explainable handling and operational interpretation of governed data
MANAGE	MANAGE 1.2	Security and Resilience	supports	Supports bounded

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				handling of sensitive or regulated data

PBSAI-F-PRV-001 — Purpose Limitation Enforcement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.2	Privacy Enhancement	equivalent	Maps lawful processing purposes and operational privacy context
GOVERN	GOVERN 1.4	Accountability	strongly-supports	Enforces governed limitations on data usage and processing
MANAGE	MANAGE 1.2	Safety	supports	Prevents unauthorized or unsafe data processing activities
MEASURE	MEASURE 3.1	Transparency	supports	Produces explainable evidence for governed data usage

PBSAI-F-PRV-002 — Lawful Basis and Regulatory Validation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.2	Accountability	equivalent	Validates contextual regulatory and lawful processing obligations
GOVERN	GOVERN 1.1	Governability	strongly-supports	Aligns operational behavior with regulatory governance requirements
MEASURE	MEASURE 3.3	Transparency	supports	Preserves explainable evidence of regulatory validation decisions
MANAGE	MANAGE 4.1	Reliability	supports	Supports adaptive governance response to regulatory risk conditions

PBSAI-F-EVID-002 — Data Lineage and Handling Traceability

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.3	Traceability	equivalent	Preserves contextual visibility into data lineage and handling

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.2	Accountability	strongly-supports	Maintains attributable operational data-governance records
MEASURE	MEASURE 3.2	Transparency	supports	Enables explainable reconstruction of governed data operations
MANAGE	MANAGE 4.1	Reliability	supports	Supports trustworthy remediation and governance adaptation

PBSAI-F-DLP-001 — Data Movement Monitoring

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.3	Transparency	equivalent	Maps operational data flows and transfer behavior
MEASURE	MEASURE 1.1	Security and Resilience	strongly-supports	Continuously monitors governed data movement conditions
MANAGE	MANAGE 1.2	Safety	supports	Prevents unsafe or unauthorized

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				data transfer operations
GOVERN	GOVERN 2.1	Accountability	supports	Preserves attributable monitoring and enforcement records

PBSAI-F-CRY-001 — Encryption at Rest and in Transit

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.4	Privacy Enhancement	equivalent	Maps cryptographic protection requirements to governed data sensitivity
GOVERN	GOVERN 3.1	Security and Resilience	strongly-supports	Enforces cryptographic governance controls
MANAGE	MANAGE 1.2	Safety	supports	Protects operational data during storage and transmission
MEASURE	MEASURE 3.1	Reliability	supports	Enables validation of cryptographic protection posture

PBSAI-F-DSP-001 — Retention Enforcement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.4	Accountability	equivalent	Preserves contextual retention and disposal obligations
GOVERN	GOVERN 1.1	Governability	strongly-supports	Enforces governed retention and deletion requirements
MANAGE	MANAGE 4.1	Reliability	supports	Supports adaptive governance response to retention-policy violations
MEASURE	MEASURE 3.2	Traceability	supports	Produces attributable retention-governance evidence

PBSAI-F-INT-001 — Data Integrity Validation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.4	Reliability	equivalent	Validates integrity and trustworthiness of governed data assets

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 2.2	Security and Resilience	strongly-supports	Continuously evaluates operational data integrity conditions
GOVERN	GOVERN 2.1	Accountability	supports	Preserves attributable validation records
MANAGE	MANAGE 1.1	Safety	supports	Prevents operational use of corrupted or untrusted data

PBSAI-F-INT-002 — Integrity Violation Detection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.4	Security and Resilience	supports	Identifies operational integrity degradation conditions
MEASURE	MEASURE 4.2	Reliability	equivalent	Detects integrity anomalies affecting governance trustworthiness
MANAGE	MANAGE 2.1	Safety	strongly-supports	Enables rapid response to integrity compromise conditions

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.3	Accountability	supports	Preserves continuous oversight of integrity-monitoring operations

6.4 MAP — Operational Risk and Dependency Context

PBSAI-A-RISK-001 — AI Risk Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 4.1	Governability	equivalent	Defines enterprise AI risk context and operational exposure
GOVERN	GOVERN 1.4	Accountability	strongly-supports	Integrates AI operational risk into governance processes
MEASURE	MEASURE 4.2	Reliability	supports	Enables evaluation of operational exposure and systemic risk
MANAGE	MANAGE 1.1	Security and Resilience	supports	Guides mitigation prioritization and treatment workflows

PBSAI-D-THREAT-001 — Threat Intelligence Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 4.2	Transparency	equivalent	Integrates evolving threat context into operational awareness
MEASURE	MEASURE 1.4	Reliability	strongly-supports	Supports adaptive operational monitoring based on threat conditions
MANAGE	MANAGE 4.1	Security and Resilience	supports	Enables adaptive governance response to changing threats
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of threat-informed governance decisions

PBSAI-D-SA-001 — Situational Awareness Correlation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 4.2	Transparency	equivalent	Correlates operational telemetry and environmental awareness

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.1	Traceability	strongly-supports	Enables explainable operational-state reconstruction
MANAGE	MANAGE 2.1	Reliability	supports	Improves incident-response and mitigation awareness
GOVERN	GOVERN 2.3	Accountability	supports	Preserves attributable situational-governance records

PBSAI-K-SCRM-001 — Supply Chain Risk Assessment

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 4.3	Transparency	equivalent	Evaluates supplier and dependency exposure relationships
GOVERN	GOVERN 4.2	Accountability	strongly-supports	Integrates supply-chain risk into governance evaluation
MEASURE	MEASURE 4.2	Reliability	supports	Assesses systemic exposure arising from dependencies

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.1	Security and Resilience	supports	Guides mitigation of supplier and dependency risk

PBSAI-K-SBOM-002 — Dependency Risk Analysis

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 4.3	Traceability	equivalent	Maps dependency exposure and component relationships
MEASURE	MEASURE 4.2	Reliability	strongly-supports	Evaluates operational exposure from vulnerable dependencies
GOVERN	GOVERN 2.2	Accountability	supports	Preserves attributable dependency-governance records
MANAGE	MANAGE 1.1	Security and Resilience	supports	Enables mitigation of dependency-related operational risk

PBSAI-I-SSE-002 — Threat Modeling and Design Analysis

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 4.4	Transparency	equivalent	Models attack paths and architectural threat exposure
GOVERN	GOVERN 1.3	Governability	strongly-supports	Aligns architectural design with operational governance objectives
MEASURE	MEASURE 4.1	Reliability	supports	Enables validation of architectural threat assumptions
MANAGE	MANAGE 1.1	Security and Resilience	supports	Guides mitigation planning for identified threat pathways

PBSAI-I-ASSUR-002 — Cross-Domain Dependency Validation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 4.4	Traceability	equivalent	Evaluates risk introduced by architectural dependencies
MEASURE	MEASURE 4.4	Reliability	strongly-supports	Correlates assurance evidence across dependent systems

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of cross-domain assurance conditions
MANAGE	MANAGE 4.2	Security and Resilience	supports	Enables adaptive mitigation of dependency-driven operational risk

6.5 MAP — Cross-Domain Contextual Governance

PBSAI operationalizes contextual governance through:

- structured context envelopes,
- provenance-aware metadata,
- identity-linked authorization,
- dependency inventories,
- trust-boundary modeling,
- telemetry correlation,
- and runtime governance binding.

The MAP alignment demonstrates that PBSAI continuously maps:

- systems,
- identities,
- privileges,
- data flows,
- dependencies,
- suppliers,

- trust boundaries,
- operational telemetry,
- and governance relationships.

Unlike static inventory-oriented governance approaches, PBSAI maintains:

- continuously evolving operational awareness,
- explainable contextual authorization,
- adaptive dependency visibility,
- provenance-linked operational state,
- and runtime contextual governance mediation.

This enables:

- trustworthy AI-enabled orchestration,
- explainable operational authorization,
- policy-aware adaptive governance,
- resilient dependency management,
- and context-aware enterprise AI security operations.

7. MEASURE Function Mapping

The MEASURE function addresses:

- monitoring,
- validation,
- testing,
- operational analytics,
- explainability,
- telemetry correlation,
- evidence generation,
- and continuous assurance evaluation.

PBSAI aligns strongly with MEASURE through:

- Domain D — Threat Intelligence, Monitoring, and Situational Awareness,
- Domain L — Analytics, AI Assurance, and Security Orchestration,
- Domain G — Incident Response and Forensics,
- Domain H — Resilience and Recovery Validation,
- and the PBSAI evidence-centric governance layer.

PBSAI operationalizes measurement through:

- continuous telemetry monitoring,
- provenance-aware validation,
- AI-output evaluation,
- drift detection,
- structured evidence generation,
- assurance analytics,
- and cross-domain operational correlation.

7.1 MEASURE — Continuous Monitoring and Operational Assessment

PBSAI-D-MON-001 — Continuous Monitoring

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.1	Reliability	equivalent	Continuously monitors AI-enabled operational environments
MEASURE	MEASURE 1.2	Security and Resilience	strongly-supports	Detects operational anomalies and governance degradation
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of operational

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				monitoring posture
MANAGE	MANAGE 2.1	Safety	supports	Enables rapid response to detected operational conditions

PBSAI-D-SA-001 — Situational Awareness Correlation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.1	Traceability	equivalent	Correlates telemetry across enterprise operational systems
MEASURE	MEASURE 4.1	Transparency	strongly-supports	Enables explainable operational-state reconstruction
MANAGE	MANAGE 2.1	Reliability	supports	Improves incident-response and mitigation awareness
GOVERN	GOVERN 2.3	Accountability	supports	Preserves attributable situational-governance records

PBSAI-L-ASSUR-001 — Continuous Assurance Evaluation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.2	Reliability	equivalent	Continuously evaluates governance and operational assurance posture
MEASURE	MEASURE 4.4	Security and Resilience	strongly-supports	Correlates assurance evidence across operational domains
GOVERN	GOVERN 4.3	Accountability	supports	Preserves continuous governance oversight
MANAGE	MANAGE 4.2	Governability	supports	Enables adaptive remediation of assurance deficiencies

PBSAI-L-ANL-001 — Governance Metrics Collection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.3	Transparency	equivalent	Collects operational governance and trustworthiness metrics
GOVERN	GOVERN 4.1	Accountability	strongly-supports	Associates metrics with

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance objectives
MANAGE	MANAGE 4.1	Reliability	supports	Supports adaptive operational improvement workflows
MEASURE	MEASURE 4.2	Traceability	supports	Preserves attributable metric lineage and evidence relationships

PBSAI-L-ANL-004 — Operational Trend and Drift Detection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.4	Reliability	equivalent	Detects operational anomalies and governance drift conditions
MEASURE	MEASURE 2.4	Security and Resilience	strongly-supports	Detects degradation in AI and governance operational posture
MANAGE	MANAGE 4.2	Governability	supports	Enables adaptive governance response to drift conditions

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.3	Accountability	supports	Preserves continuous oversight of operational trustworthiness

PBSAI-D-THREAT-003 — Threat-Adaptive Monitoring

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.4	Reliability	equivalent	Dynamically adjusts monitoring based on operational threat conditions
MAP	MAP 4.2	Transparency	strongly-supports	Integrates evolving threat context into operational awareness
MANAGE	MANAGE 4.1	Security and Resilience	supports	Enables adaptive governance response to changing threats
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of threat-adaptive governance decisions

PBSAI-H-TEST-001 — Continuity and Recovery Testing

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.5	Reliability	equivalent	Validates resilience and recovery readiness
MANAGE	MANAGE 3.5	Security and Resilience	strongly-supports	Supports resilient operational continuity validation
GOVERN	GOVERN 4.4	Accountability	supports	Preserves attributable resilience-testing governance evidence
MEASURE	MEASURE 5.2	Transparency	supports	Enables explainable recovery-readiness evaluation

PBSAI-I-ASSUR-003 — Continuous Architectural Monitoring

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.2	Reliability	equivalent	Continuously evaluates architectural conformance and operational drift

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 1.3	Governability	strongly-supports	Preserves governed architectural oversight
MEASURE	MEASURE 4.1	Traceability	supports	Enables correlation of telemetry with architectural controls
MANAGE	MANAGE 1.1	Security and Resilience	supports	Supports mitigation of architecture-driven operational exposure

7.2 MEASURE — AI Validation, Explainability, and Trustworthiness

PBSAI-L-AI-001 — AI Model Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 2.1	Governability	equivalent	Applies governed oversight to AI operational systems
GOVERN	GOVERN 1.5	Accountability	strongly-supports	Aligns AI operational behavior with enterprise governance requirements
MANAGE	MANAGE 4.2	Reliability	supports	Enables adaptive

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance response to AI operational findings
MAP	MAP 4.1	Transparency	supports	Associates AI operational behavior with contextual risk awareness

PBSAI-L-AI-002 — AI Output Validation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 2.2	Reliability	equivalent	Validates AI-generated outputs before operational acceptance
MEASURE	MEASURE 2.3	Explainability	strongly-supports	Preserves explainable validation semantics
MANAGE	MANAGE 1.1	Safety	supports	Prevents unsafe operational actions derived from invalid outputs
GOVERN	GOVERN 2.3	Transparency	supports	Enables explainable governance review of AI

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				operational behavior

PBSAI-L-AI-003 — AI Confidence and Explainability Enforcement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 2.3	Explainability	equivalent	Preserves confidence scoring and rationale metadata
GOVERN	GOVERN 2.3	Transparency	strongly-supports	Enables explainable operational interpretation
MEASURE	MEASURE 2.4	Reliability	supports	Detects anomalous AI confidence and behavioral conditions
MANAGE	MANAGE 4.2	Governability	supports	Enables adaptive governance informed by explainability evaluation

PBSAI-L-AI-004 — Human Oversight for High-Impact AI Decisions

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 2.5	Human Oversight	equivalent	Requires governed human authorization for high-impact operational actions
GOVERN	GOVERN 1.6	Accountability	strongly-supports	Preserves attributable approval and escalation workflows
MANAGE	MANAGE 5.1	Safety	supports	Prevents unsafe autonomous operational execution
MAP	MAP 2.4	Transparency	supports	Associates stakeholder authority with governed AI decision workflows

PBSAI-L-AI-005 — AI Drift and Behavior Monitoring

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 2.4	Reliability	equivalent	Detects AI behavioral drift and anomalous operational conditions
MEASURE	MEASURE 1.1	Security and Resilience	strongly-supports	Continuously monitors

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				operational AI trustworthiness
MANAGE	MANAGE 4.2	Governability	supports	Enables adaptive governance response to AI drift
GOVERN	GOVERN 4.3	Accountability	supports	Preserves continuous oversight of AI operational behavior

PBSAI-L-AI-006 — AI Assurance Evidence and Traceability

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 2.5	Traceability	equivalent	Produces attributable evidence for AI operational behavior
MEASURE	MEASURE 3.3	Transparency	strongly-supports	Enables explainable AI operational reconstruction
GOVERN	GOVERN 2.1	Accountability	supports	Preserves attributable AI governance evidence
MANAGE	MANAGE 4.2	Governability	supports	Supports adaptive AI assurance and

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance workflows

PBSAI-GEN-OC-001 — Structured Output Contracts

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 2.3	Explainability	equivalent	Standardizes explainable machine-readable operational outputs
MEASURE	MEASURE 3.1	Traceability	strongly-supports	Produces structured evidence-compatible output artifacts
GOVERN	GOVERN 2.3	Transparency	supports	Enables explainable governance review of AI operational behavior
MANAGE	MANAGE 5.3	Safety	supports	Constrains operational execution through deterministic schema validation

PBSAI-GEN-MCP-001 — Context Envelope Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 2.2	Transparency	equivalent	Enables explainable contextual evaluation and operational interpretation
MAP	MAP 2.3	Traceability	strongly-supports	Preserves contextual metadata and governance bindings
GOVERN	GOVERN 2.2	Accountability	supports	Associates operational actions with contextual governance state
MANAGE	MANAGE 5.4	Governability	supports	Preserves governance constraints during operational execution

PBSAI-GEN-EVID-001 — Evidence Generation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 3.1	Traceability	equivalent	Generates structured operational telemetry and audit evidence

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 3.3	Transparency	strongly-supports	Enables explainable operational reconstruction and review
GOVERN	GOVERN 2.1	Accountability	supports	Produces attributable governance evidence
MANAGE	MANAGE 4.1	Reliability	supports	Supports adaptive governance improvement through operational evidence

PBSAI-GEN-PROV-001 — Provenance and Attribution

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 3.2	Traceability	equivalent	Preserves provenance-linked operational history
GOVERN	GOVERN 2.2	Accountability	strongly-supports	Maintains attributable provenance for operational actions
MAP	MAP 4.3	Transparency	supports	Enables dependency and

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				provenance visibility
MANAGE	MANAGE 2.4	Reliability	supports	Supports trustworthy remediation and forensic reconstruction

PBSAI-D-EVID-001 — Security Event Logging

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 3.1	Traceability	equivalent	Produces telemetry evidence for operational analysis
MEASURE	MEASURE 4.1	Transparency	strongly-supports	Enables explainable reconstruction of operational events
GOVERN	GOVERN 2.1	Accountability	supports	Preserves attributable security-event records
MANAGE	MANAGE 2.1	Reliability	supports	Supports incident-response and mitigation workflows

PBSAI-F-EVID-001 — Data Action Evidence Records

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 3.2	Traceability	equivalent	Preserves attributable records for governed data operations
GOVERN	GOVERN 2.1	Accountability	strongly-supports	Associates operational data actions with governance obligations
MAP	MAP 3.3	Transparency	supports	Enables explainable reconstruction of data handling workflows
MANAGE	MANAGE 4.1	Reliability	supports	Supports adaptive remediation of governed data incidents

7.3 MEASURE — Evidence Generation, Provenance, and Cross-Domain Correlation

PBSAI-G-RCA-002 — Incident Correlation and Timeline Reconstruction

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.1	Traceability	equivalent	Reconstructs operational sequences and incident timelines

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 3.3	Transparency	strongly-supports	Enables explainable forensic and operational reconstruction
GOVERN	GOVERN 2.1	Accountability	supports	Preserves attributable operational history
MANAGE	MANAGE 2.4	Reliability	supports	Supports trustworthy remediation and incident analysis

PBSAI-L-ANL-002 — Cross-Domain Telemetry Correlation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.1	Traceability	equivalent	Correlates telemetry and operational evidence across governance domains
MEASURE	MEASURE 4.2	Reliability	strongly-supports	Enables cross-domain operational analytics and anomaly detection
GOVERN	GOVERN 2.4	Transparency	supports	Supports operational transparency through

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				telemetry aggregation
MANAGE	MANAGE 4.1	Governability	supports	Enables adaptive governance informed by operational telemetry

PBSAI-D-SA-002 — Behavioral and Operational Correlation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.1	Transparency	equivalent	Detects abnormal operational behavior patterns
MEASURE	MEASURE 4.2	Reliability	strongly-supports	Correlates behavioral indicators with operational anomalies
MANAGE	MANAGE 2.1	Security and Resilience	supports	Enables rapid mitigation of abnormal operational conditions
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of behavioral-governance analytics

PBSAI-L-ANL-003 — Risk and Exposure Analytics

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.2	Reliability	equivalent	Produces operational exposure and governance risk analytics
MAP	MAP 4.1	Transparency	strongly-supports	Associates exposure analytics with contextual operational risk
GOVERN	GOVERN 4.2	Accountability	supports	Preserves attributable risk-analysis governance records
MANAGE	MANAGE 1.1	Security and Resilience	supports	Supports mitigation prioritization and operational treatment planning

PBSAI-F-INT-002 — Integrity Violation Detection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.2	Security and Resilience	equivalent	Detects integrity anomalies affecting governance trustworthiness

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.4	Reliability	strongly-supports	Identifies operational integrity degradation conditions
MANAGE	MANAGE 2.1	Safety	supports	Enables rapid response to integrity compromise conditions
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of integrity-monitoring operations

PBSAI-J-SURV-003 — Cross-Domain Event Correlation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.1	Traceability	equivalent	Correlates cyber, physical, and environmental operational events
MEASURE	MEASURE 4.2	Reliability	strongly-supports	Enables detection of multi-domain operational anomalies
GOVERN	GOVERN 2.4	Transparency	supports	Preserves explainable cross-domain

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.1	Security and Resilience	supports	situational awareness Improves coordinated response to compound operational events

PBSAI-K-ASSUR-003 — Continuous Assurance Monitoring

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.4	Reliability	equivalent	Continuously evaluates supply-chain assurance posture
GOVERN	GOVERN 4.3	Accountability	strongly-supports	Preserves oversight of supplier and dependency assurance conditions
MAP	MAP 4.3	Transparency	supports	Associates assurance posture with dependency visibility
MANAGE	MANAGE 1.1	Security and Resilience	supports	Enables adaptive mitigation of supply-chain

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				operational exposure

PBSAI-L-ASSUR-002 — Cross-Domain Assurance Correlation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.4	Reliability	equivalent	Correlates assurance evidence across operational domains
MEASURE	MEASURE 4.1	Traceability	strongly- supports	Links assurance telemetry across governance workflows
GOVERN	GOVERN 4.3	Accountability	supports	Preserves attributable governance assurance records
MANAGE	MANAGE 4.2	Governability	supports	Enables adaptive governance response to assurance degradation

PBSAI-I-ASSUR-002 — Cross-Domain Dependency Validation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.4	Reliability	equivalent	Correlates assurance evidence across dependent systems
MAP	MAP 4.4	Traceability	strongly-supports	Evaluates risk introduced by architectural dependencies
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of cross-domain assurance conditions
MANAGE	MANAGE 4.2	Security and Resilience	supports	Enables adaptive mitigation of dependency-driven operational risk

7.4 MEASURE — Continuous Validation and Adaptive Assurance

PBSAI-L-FEED-001 — Operational Feedback Collection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 5.1	Transparency	equivalent	Preserves operational feedback for governance evaluation
MANAGE	MANAGE 4.1	Reliability	strongly-supports	Supports adaptive operational

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				improvement workflows
GOVERN	GOVERN 4.4	Accountability	supports	Collects operational findings and governance outcomes
MANAGE	MANAGE 4.3	Governability	supports	Enables continuously adaptive governance processes

PBSAI-L-FEED-003 — Control Improvement Recommendation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 5.2	Transparency	equivalent	Preserves explainable governance-improvement rationale
GOVERN	GOVERN 4.4	Governability	strongly-supports	Recommends governance improvements using operational evidence
MANAGE	MANAGE 4.2	Reliability	supports	Enables adaptive mitigation and operational optimization
MEASURE	MEASURE 4.2	Accountability	supports	Associates recommendations

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				with operational evidence and findings

PBSAI-H-TEST-005 — Improvement and Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 5.2	Accountability	equivalent	Integrates testing evidence into governance evaluation workflows
MANAGE	MANAGE 3.5	Security and Resilience	strongly-supports	Continuously improves resilience and recovery readiness
GOVERN	GOVERN 4.4	Reliability	supports	Uses resilience-testing outcomes to improve governance posture
MANAGE	MANAGE 4.3	Governability	supports	Enables adaptive continuity and resilience governance improvement

PBSAI-G-RCA-004 — Lessons Learned and Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 5.1	Transparency	equivalent	Preserves operational lessons learned for governance improvement
MANAGE	MANAGE 4.1	Reliability	strongly-supports	Integrates incident findings into operational improvement workflows
GOVERN	GOVERN 4.4	Accountability	supports	Maintains attributable governance-improvement evidence
MEASURE	MEASURE 4.1	Traceability	supports	Enables explainable linkage between incidents and governance adaptation

PBSAI-L-FEED-004 — Governance Adaptation and Optimization

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 5.3	Governability	equivalent	Continuously adapts governance workflows and operational controls
MANAGE	MANAGE 4.3	Reliability	strongly-supports	Enables resilient and

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				adaptive governance evolution
GOVERN	GOVERN 4.4	Accountability	supports	Preserves attributable governance adaptation decisions
MANAGE	MANAGE 4.4	Security and Resilience	supports	Supports resilience-oriented governance optimization

PBSAI-L-ASSUR-004 — Assurance Escalation and Remediation Coordination

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 5.3	Reliability	equivalent	Coordinates remediation for governance and assurance deficiencies
MANAGE	MANAGE 4.2	Governability	strongly-supports	Enables adaptive remediation of assurance degradation
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of remediation coordination activities

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.4	Transparency	supports	Enables explainable assurance escalation workflows

PBSAI-D-THREAT-005 — Threat Intelligence Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 5.2	Reliability	equivalent	Integrates evolving threat intelligence into operational validation
MAP	MAP 4.2	Transparency	strongly-supports	Associates operational adaptation with evolving threat context
MANAGE	MANAGE 4.1	Security and Resilience	supports	Enables adaptive governance response to threat evolution
GOVERN	GOVERN 4.4	Accountability	supports	Preserves attributable threat-adaptation governance records

PBSAI-GEN-FEED-001 — Continuous Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 5.3	Governability	equivalent	Operationalizes adaptive governance and continuous improvement
MANAGE	MANAGE 4.4	Reliability	strongly-supports	Enables continuously adaptive operational governance
GOVERN	GOVERN 4.4	Accountability	supports	Preserves attributable governance adaptation workflows
MEASURE	MEASURE 4.1	Transparency	supports	Enables explainable operational improvement and optimization

7.5 MEASURE — Cross-Domain Measurement Governance

PBSAI operationalizes measurement as:

- continuous operational validation,
- telemetry fusion,
- provenance-aware assurance,
- explainable governance analytics,
- evidence-centric accountability,
- and adaptive operational trustworthiness evaluation.

The MEASURE alignment demonstrates that PBSAI continuously measures:

- operational trustworthiness,

- AI behavior,
- governance effectiveness,
- resilience posture,
- dependency exposure,
- data integrity,
- supply-chain assurance,
- and runtime policy conformance.

Unlike static assessment-oriented governance approaches, PBSAI maintains:

- continuously adaptive monitoring,
- runtime explainability,
- cross-domain telemetry fusion,
- operational evidence linkage,
- and continuously evolving assurance analytics.

This enables:

- trustworthy AI-enabled enterprise operations,
- explainable governance validation,
- resilient operational assurance,
- evidence-centric accountability,
- and adaptive enterprise AI governance evaluation.

8. MANAGE Function Mapping

The MANAGE function addresses:

- operational mitigation,
- incident response,
- resilience,
- recovery,
- adaptive governance,

- bounded remediation,
- orchestration governance,
- and continuous operational improvement.

PBSAI aligns strongly with MANAGE through:

- Domain E — Protective Enforcement and Response,
- Domain G — Incident Response and Forensics,
- Domain H — Resilience, Continuity, and Recovery,
- Domain L — Security Orchestration and AI Assurance,
- and the PBSAI runtime governance control layer.

PBSAI operationalizes management through:

- deterministic authorization mediation,
- bounded remediation workflows,
- rollback semantics,
- explainable orchestration,
- resilience-aware recovery,
- and continuously adaptive but policy-constrained operational governance.

8.1 MANAGE — Risk Mitigation and Protective Enforcement

PBSAI-E-PRO-001 — Preventive Control Definition

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.1	Safety	equivalent	Establishes operational protective controls and policy enforcement
GOVERN	GOVERN 3.1	Governability	strongly-supports	Enforces policy-bound

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				operational execution
MEASURE	MEASURE 4.1	Reliability	supports	Enables validation of preventive-control effectiveness
MAP	MAP 1.3	Transparency	supports	Associates controls with trust-boundary and segmentation awareness

PBSAI-E-PRO-002 — Segmentation and Isolation Enforcement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.1	Security and Resilience	equivalent	Limits operational exposure through segmentation and isolation
GOVERN	GOVERN 3.1	Safety	strongly-supports	Enforces bounded operational communication pathways
MEASURE	MEASURE 4.1	Reliability	supports	Enables validation of segmentation effectiveness

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 1.3	Transparency	supports	Preserves trust-boundary visibility during operational enforcement

PBSAI-E-PRO-003 — Endpoint and Workload Protection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.2	Security and Resilience	equivalent	Applies runtime protection to enterprise systems and workloads
MEASURE	MEASURE 1.1	Reliability	strongly-supports	Continuously monitors protected operational environments
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of endpoint-protection governance posture
MAP	MAP 1.1	Transparency	supports	Associates operational protection with governed assets

PBSAI-E-PRO-004 — Protected Data Access Enforcement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.2	Privacy Enhancement	equivalent	Restricts unauthorized access to governed data resources
GOVERN	GOVERN 3.1	Accountability	strongly-supports	Enforces governed data-access authorization policies
MEASURE	MEASURE 3.2	Traceability	supports	Produces attributable data-access evidence
MAP	MAP 3.1	Transparency	supports	Associates protection requirements with classified data assets

PBSAI-E-ENF-003 — Response Scope Constraint

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.3	Safety	equivalent	Constrains operational remediation and enforcement scope
GOVERN	GOVERN 3.1	Governability	strongly-supports	Prevents operational actions from exceeding

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				approved boundaries
MAP	MAP 1.3	Transparency	supports	Preserves trust-boundary and segmentation awareness
MEASURE	MEASURE 4.2	Reliability	supports	Enables validation of bounded remediation behavior

PBSAI-GEN-SAFE-001 — Safe Execution Constraint

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.3	Reliability	equivalent	Prevents unsafe or uncontrolled operational behavior
GOVERN	GOVERN 3.1	Safety	strongly-supports	Enforces bounded operational execution semantics
MANAGE	MANAGE 5.3	Governability	supports	Constrains orchestration workflows within approved operational limits
MEASURE	MEASURE 1.2	Security and Resilience	supports	Enables validation of

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				operational safety conditions

PBSAI-GEN-SAFE-003 — Kill Switch / Emergency Stop

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.4	Safety	equivalent	Enables rapid interruption of unsafe operational behavior
GOVERN	GOVERN 3.1	Governability	strongly-supports	Maintains deterministic override authority
MEASURE	MEASURE 1.1	Reliability	supports	Enables validation of emergency interruption capability
MANAGE	MANAGE 2.2	Security and Resilience	supports	Supports containment during compromise or operational instability

PBSAI-I-SEG-004 — Dynamic Segmentation Enforcement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.1	Security and Resilience	equivalent	Dynamically adapts segmentation protections during evolving threats
MAP	MAP 1.3	Transparency	strongly-supports	Preserves trust-boundary visibility during adaptive segmentation
MEASURE	MEASURE 4.2	Reliability	supports	Enables validation of adaptive segmentation effectiveness
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of dynamic enforcement activities

8.2 MANAGE — Incident Response and Containment

PBSAI-G-IR-001 — Incident Identification

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.1	Reliability	equivalent	Detects and identifies operational incidents requiring response

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.1	Security and Resilience	strongly-supports	Continuously monitors for anomalous operational conditions
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of incident-detection governance posture
MAP	MAP 4.2	Transparency	supports	Associates operational incidents with contextual threat awareness

PBSAI-G-IR-003 — Incident Triage and Routing

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.1	Governability	equivalent	Coordinates response routing and escalation workflows
GOVERN	GOVERN 1.6	Accountability	strongly-supports	Establishes governed escalation and operational review pathways

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 4.1	Traceability	supports	Preserves attributable incident-routing records
MAP	MAP 2.4	Transparency	supports	Associates stakeholders and response authorities with incidents

PBSAI-G-CONT-001 — Incident Containment Execution

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.2	Security and Resilience	equivalent	Executes bounded operational containment actions
GOVERN	GOVERN 3.1	Safety	strongly-supports	Maintains governed containment execution authority
MEASURE	MEASURE 1.1	Reliability	supports	Enables validation of containment effectiveness
MAP	MAP 4.2	Transparency	supports	Preserves contextual awareness during containment workflows

PBSAI-G-CONT-002 — Dynamic Isolation and Segmentation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.2	Security and Resilience	equivalent	Restricts propagation during operational compromise
MAP	MAP 1.3	Transparency	strongly-supports	Preserves segmentation and trust-boundary awareness
MEASURE	MEASURE 4.1	Reliability	supports	Enables validation of dynamic isolation effectiveness
GOVERN	GOVERN 3.1	Safety	supports	Maintains bounded containment governance controls

PBSAI-G-CONT-004 — Emergency Containment Override

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.3	Safety	equivalent	Supports emergency interruption of unsafe operational conditions

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.1	Governability	strongly-supports	Maintains deterministic override authority during crisis conditions
MEASURE	MEASURE 1.5	Reliability	supports	Enables validation of emergency response readiness
MAP	MAP 4.2	Transparency	supports	Associates emergency actions with operational threat context

PBSAI-G-FOR-001 — Forensic Evidence Collection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.4	Traceability	equivalent	Preserves operational evidence during incident handling
MEASURE	MEASURE 3.1	Transparency	strongly-supports	Enables explainable forensic reconstruction
GOVERN	GOVERN 2.1	Accountability	supports	Maintains attributable forensic evidence records

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 3.3	Reliability	supports	Associates forensic evidence with governed operational context

PBSAI-G-RCA-001 — Root Cause Identification

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.5	Reliability	equivalent	Determines systemic causes of operational failures or incidents
MEASURE	MEASURE 4.2	Transparency	strongly-supports	Enables explainable operational-failure analysis
GOVERN	GOVERN 4.4	Accountability	supports	Preserves attributable governance-improvement evidence
MAP	MAP 4.1	Traceability	supports	Associates root causes with contextual operational risk conditions

PBSAI-G-RCA-004 — Lessons Learned and Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.5	Governability	equivalent	Integrates incident findings into operational improvement workflows
MEASURE	MEASURE 5.1	Transparency	strongly-supports	Preserves explainable lessons-learned governance evidence
GOVERN	GOVERN 4.4	Accountability	supports	Maintains attributable governance-improvement records
MANAGE	MANAGE 4.3	Reliability	supports	Enables continuously adaptive operational governance improvement

PBSAI-G-EVID-001 — Incident Evidence Generation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.4	Traceability	equivalent	Produces evidence for incident and response workflows
MEASURE	MEASURE 3.1	Transparency	strongly-supports	Enables explainable reconstruction of

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 2.1	Accountability	supports	incident activities Preserves attributable operational response records
MANAGE	MANAGE 4.1	Reliability	supports	Supports adaptive incident-response improvement workflows

PBSAI-G-EVID-003 — Response Attribution and Accountability

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 2.4	Accountability	equivalent	Maintains attributable operational response history
MEASURE	MEASURE 3.2	Traceability	strongly-supports	Preserves forensic linkage between actions and actors
GOVERN	GOVERN 2.1	Transparency	supports	Enables explainable governance review of incident-

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				response activities
MAP	MAP 2.4	Reliability	supports	Associates response activities with governed stakeholder authority

PBSAI-E-ENF-002 — Response Authorization Validation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.1	Human Oversight	equivalent	Requires governed authorization prior to operational response actions
GOVERN	GOVERN 3.1	Accountability	strongly-supports	Enforces policy-bound response authorization workflows
MEASURE	MEASURE 3.2	Traceability	supports	Produces attributable response authorization evidence
MAP	MAP 2.2	Transparency	supports	Associates operational response authority with contextual

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				authorization state

8.3 MANAGE — Resilience, Recovery, and Continuity

PBSAI-H-RES-001 — Operational Resilience Planning

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 3.1	Security and Resilience	equivalent	Defines enterprise resilience and continuity governance
GOVERN	GOVERN 4.1	Accountability	strongly- supports	Aligns resilience planning with enterprise governance objectives
MAP	MAP 4.1	Transparency	supports	Associates resilience posture with contextual operational risk
MEASURE	MEASURE 1.5	Reliability	supports	Enables validation of resilience readiness and continuity posture

PBSAI-H-BKP-001 — Backup Definition and Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 3.2	Reliability	equivalent	Governs backup creation and recoverability requirements
GOVERN	GOVERN 3.2	Accountability	strongly-supports	Establishes governed recovery and backup oversight
MEASURE	MEASURE 1.5	Security and Resilience	supports	Enables validation of recoverability and backup integrity
MAP	MAP 3.4	Transparency	supports	Associates retention and recovery requirements with governed assets

PBSAI-H-REC-001 — Recovery Planning

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 3.3	Reliability	equivalent	Defines governed recovery and restoration workflows
GOVERN	GOVERN 4.1	Accountability	strongly-supports	Aligns recovery planning with enterprise

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance objectives
MAP	MAP 4.1	Transparency	supports	Associates recovery workflows with contextual operational dependencies
MEASURE	MEASURE 1.5	Security and Resilience	supports	Enables validation of operational recovery readiness

PBSAI-H-REC-003 — Recovery Execution

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 3.3	Security and Resilience	equivalent	Executes operational recovery and restoration activities
GOVERN	GOVERN 3.2	Accountability	strongly-supports	Maintains governed oversight of recovery execution
MEASURE	MEASURE 1.5	Reliability	supports	Enables validation of restoration effectiveness

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MAP	MAP 4.4	Transparency	supports	Associates recovery operations with dependency and trust-boundary awareness

PBSAI-H-REC-006 — Recovery Rollback and Safe Restoration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 3.4	Reliability	equivalent	Supports controlled rollback and operational restoration
GOVERN	GOVERN 3.2	Safety	strongly-supports	Maintains safe operational restoration behavior
MANAGE	MANAGE 5.3	Security and Resilience	supports	Preserves resilient continuity during recovery operations
MEASURE	MEASURE 1.5	Reliability	supports	Enables validation of restoration and rollback readiness

PBSAI-H-FAIL-001 — Failover Orchestration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 3.4	Security and Resilience	equivalent	Maintains continuity during operational disruption
MAP	MAP 4.4	Transparency	strongly-supports	Associates failover operations with architectural dependencies
MEASURE	MEASURE 1.5	Reliability	supports	Enables validation of continuity and failover readiness
GOVERN	GOVERN 4.1	Accountability	supports	Preserves oversight of failover governance operations

PBSAI-H-TEST-001 — Continuity and Recovery Testing

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 3.5	Security and Resilience	equivalent	Validates resilience and continuity readiness
MEASURE	MEASURE 1.5	Reliability	strongly-supports	Continuously evaluates recovery readiness and operational resilience

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.4	Accountability	supports	Preserves attributable resilience-testing evidence
MAP	MAP 4.1	Transparency	supports	Associates resilience-testing activities with operational risk context

PBSAI-H-TEST-005 — Improvement and Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 3.5	Reliability	equivalent	Continuously improves resilience and recovery readiness
MEASURE	MEASURE 5.2	Accountability	strongly-supports	Integrates testing evidence into governance evaluation workflows
GOVERN	GOVERN 4.4	Transparency	supports	Preserves explainable governance-improvement rationale
MANAGE	MANAGE 4.3	Governability	supports	Enables adaptive continuity and resilience

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance improvement

PBSAI-H-EVID-001 — Resilience Event Logging

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 3.3	Traceability	equivalent	Captures evidence for resilience and recovery operations
MEASURE	MEASURE 3.1	Transparency	strongly-supports	Enables explainable reconstruction of continuity operations
GOVERN	GOVERN 2.1	Accountability	supports	Preserves attributable resilience-governance records
MAP	MAP 4.1	Reliability	supports	Associates continuity events with operational dependencies and recovery context

PBSAI-GEN-SAFE-002 — Rollback Requirement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.3	Security and Resilience	equivalent	Preserves resilient operational continuity during failure conditions
GOVERN	GOVERN 3.2	Safety	strongly-supports	Enables recovery from unsafe operational outcomes
MEASURE	MEASURE 1.5	Reliability	supports	Enables validation of rollback readiness and restoration capability
MAP	MAP 4.4	Transparency	supports	Associates rollback workflows with dependency and operational context

PBSAI-GEN-SAFE-003 — Kill Switch / Emergency Stop

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 1.4	Safety	equivalent	Enables rapid interruption of unsafe operational behavior

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.1	Governability	strongly-supports	Maintains deterministic override authority
MEASURE	MEASURE 1.1	Reliability	supports	Enables validation of emergency interruption capability
MANAGE	MANAGE 2.2	Security and Resilience	supports	Supports containment during compromise or operational instability

8.4 MANAGE — Adaptive Governance and Continuous Improvement

PBSAI-L-FEED-001 — Operational Feedback Collection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.1	Reliability	equivalent	Collects operational findings and governance outcomes for adaptive improvement
MEASURE	MEASURE 5.1	Transparency	strongly-supports	Preserves explainable operational feedback evidence

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.4	Accountability	supports	Maintains attributable governance-improvement records
MAP	MAP 4.1	Traceability	supports	Associates operational findings with contextual risk conditions

PBSAI-L-FEED-003 — Control Improvement Recommendation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.2	Governability	equivalent	Recommends governance improvements using operational evidence
MEASURE	MEASURE 5.2	Transparency	strongly-supports	Preserves explainable governance-improvement rationale
GOVERN	GOVERN 4.4	Accountability	supports	Associates governance improvements with operational findings
MEASURE	MEASURE 4.2	Reliability	supports	Enables adaptive optimization

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				using operational analytics

PBSAI-L-FEED-004 — Governance Adaptation and Optimization

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.3	Governability	equivalent	Continuously adapts governance workflows and operational controls
GOVERN	GOVERN 4.4	Accountability	strongly-supports	Preserves attributable governance adaptation decisions
MEASURE	MEASURE 5.3	Reliability	supports	Enables adaptive operational optimization
MANAGE	MANAGE 4.4	Security and Resilience	supports	Supports resilience-oriented governance evolution

PBSAI-L-ASSUR-003 — Assurance Drift Detection

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.2	Reliability	equivalent	Detects degradation in governance and operational assurance posture
MEASURE	MEASURE 2.4	Security and Resilience	strongly-supports	Continuously monitors assurance integrity and operational drift
GOVERN	GOVERN 4.3	Accountability	supports	Preserves oversight of assurance-monitoring activities
MEASURE	MEASURE 4.4	Transparency	supports	Enables explainable assurance degradation analysis

PBSAI-L-ASSUR-004 — Assurance Escalation and Remediation Coordination

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.2	Governability	equivalent	Coordinates remediation for governance and assurance deficiencies
MEASURE	MEASURE 5.3	Reliability	strongly-supports	Enables adaptive remediation of

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 4.3	Accountability	supports	assurance degradation Preserves attributable remediation-coordination evidence
MAP	MAP 4.4	Transparency	supports	Associates remediation workflows with contextual dependency awareness

PBSAI-D-THREAT-005 — Threat Intelligence Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.1	Security and Resilience	equivalent	Integrates evolving threat intelligence into operational governance
MAP	MAP 4.2	Transparency	strongly-supports	Associates operational adaptation with evolving threat context
MEASURE	MEASURE 5.2	Reliability	supports	Enables adaptive operational validation informed by

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				threat intelligence
GOVERN	GOVERN 4.4	Accountability	supports	Preserves attributable threat-adaptation governance records

PBSAI-L-ORCH-001 — Security Workflow Orchestration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.4	Governability	equivalent	Coordinates adaptive operational governance workflows
GOVERN	GOVERN 3.1	Accountability	strongly-supports	Preserves governed orchestration authority and workflow control
MEASURE	MEASURE 4.1	Traceability	supports	Produces attributable orchestration telemetry and workflow evidence
MAP	MAP 2.3	Transparency	supports	Associates orchestration activities with contextual

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance state

PBSAI-GEN-FEED-001 — Continuous Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.4	Governability	equivalent	Operationalizes adaptive governance and continuous improvement
MEASURE	MEASURE 5.3	Reliability	strongly-supports	Enables continuously adaptive operational governance
GOVERN	GOVERN 4.4	Accountability	supports	Preserves attributable governance adaptation workflows
MEASURE	MEASURE 4.1	Transparency	supports	Enables explainable operational optimization and governance evolution

PBSAI-L-ANL-003 — Risk and Exposure Analytics

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.1	Reliability	equivalent	Produces operational exposure and governance risk analytics
MAP	MAP 4.1	Transparency	strongly-supports	Associates exposure analytics with contextual operational risk
GOVERN	GOVERN 4.2	Accountability	supports	Preserves attributable governance-risk analysis records
MEASURE	MEASURE 4.2	Security and Resilience	supports	Enables adaptive operational mitigation planning

PBSAI-G-RCA-004 — Lessons Learned and Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.3	Reliability	equivalent	Integrates incident findings into operational improvement workflows
MEASURE	MEASURE 5.1	Transparency	strongly-supports	Preserves explainable lessons-learned

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance evidence
GOVERN	GOVERN 4.4	Accountability	supports	Maintains attributable governance-improvement records
MAP	MAP 4.1	Traceability	supports	Associates operational improvements with incident context and operational risk

PBSAI-H-TEST-005 — Improvement and Feedback Integration

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.3	Security and Resilience	equivalent	Continuously improves resilience and recovery readiness
MEASURE	MEASURE 5.2	Accountability	strongly-supports	Integrates resilience-testing evidence into governance workflows
GOVERN	GOVERN 4.4	Reliability	supports	Uses resilience outcomes to improve governance posture

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 4.4	Governability	supports	Enables adaptive continuity and resilience governance improvement

8.5 MANAGE — Human Oversight and Operational Governance Constraints

PBSAI-L-AI-004 — Human Oversight for High-Impact AI Decisions

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.1	Human Oversight	equivalent	Requires human authorization for high-impact operational actions
GOVERN	GOVERN 1.6	Accountability	strongly-supports	Preserves attributable approval and escalation workflows
MEASURE	MEASURE 2.5	Transparency	supports	Enables explainable review of governed AI operational behavior
MAP	MAP 2.4	Governability	supports	Associates stakeholder authority with operational decision context

PBSAI-GEN-HITL-001 — Human Approval Requirement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.1	Safety	equivalent	Prevents unsafe autonomous operational execution
GOVERN	GOVERN 1.6	Human Oversight	strongly-supports	Requires governed authorization prior to operational execution
MEASURE	MEASURE 3.2	Accountability	supports	Produces attributable authorization evidence
MAP	MAP 2.4	Transparency	supports	Associates stakeholders and authorization responsibilities with execution context

PBSAI-E-ENF-002 — Response Authorization Validation

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.2	Accountability	equivalent	Validates authorization before remediation or response execution

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
GOVERN	GOVERN 3.1	Governability	strongly-supports	Enforces policy-bound response authorization workflows
MEASURE	MEASURE 3.2	Traceability	supports	Produces attributable response authorization evidence
MAP	MAP 2.2	Transparency	supports	Associates operational response authority with contextual authorization state

PBSAI-I-DEP-003 — Deployment Scope Constraints

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.2	Safety	equivalent	Restricts deployment authority and operational scope
GOVERN	GOVERN 3.2	Governability	strongly-supports	Prevents unauthorized or unsafe deployment execution
MEASURE	MEASURE 3.4	Traceability	supports	Enables attributable deployment

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance records
MAP	MAP 1.2	Transparency	supports	Preserves deployment-context visibility

PBSAI-G-CONT-003 — Containment Scope Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.2	Governability	equivalent	Prevents containment actions from exceeding approved boundaries
GOVERN	GOVERN 3.1	Safety	strongly-supports	Maintains bounded containment governance controls
MAP	MAP 1.3	Transparency	supports	Preserves trust-boundary visibility during containment
MEASURE	MEASURE 4.1	Reliability	supports	Enables validation of governed containment behavior

PBSAI-L-ORCH-004 — Autonomous Workflow Constraint Enforcement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.3	Safety	equivalent	Prevents unsafe autonomous operational execution
GOVERN	GOVERN 3.1	Governability	strongly-supports	Constrains orchestration workflows within policy-defined boundaries
MEASURE	MEASURE 4.1	Reliability	supports	Enables validation of orchestration governance behavior
MAP	MAP 2.3	Transparency	supports	Preserves contextual governance constraints during orchestration

PBSAI-GEN-SAFE-002 — Rollback Requirement

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.3	Security and Resilience	equivalent	Preserves resilient operational continuity during failure conditions
GOVERN	GOVERN 3.2	Safety	strongly-supports	Enables recovery from unsafe

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MEASURE	MEASURE 1.5	Reliability	supports	operational outcomes Enables validation of rollback readiness and restoration capability
MAP	MAP 4.4	Transparency	supports	Associates rollback workflows with dependency and operational context

PBSAI-GEN-MCP-001 — Context Envelope Governance

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
MANAGE	MANAGE 5.4	Governability	equivalent	Preserves governance constraints during operational execution
MAP	MAP 2.3	Traceability	strongly-supports	Preserves contextual metadata and governance bindings
GOVERN	GOVERN 2.2	Accountability	supports	Associates operational actions with contextual

AI RMF Function	AI RMF Category/Subcategory	Trustworthiness Characteristic	Relationship	Rationale
				governance state
MEASURE	MEASURE 2.2	Transparency	supports	Enables explainable contextual evaluation and operational interpretation

8.6 MANAGE — Cross-Domain Operational Governance

PBSAI operationalizes management through:

- bounded remediation,
- deterministic governance mediation,
- explainable orchestration,
- resilience-aware recovery,
- continuous operational validation,
- and adaptive but policy-constrained execution.

The MANAGE alignment demonstrates that PBSAI continuously governs:

- operational mitigation,
- response coordination,
- recovery behavior,
- resilience posture,
- adaptive governance,
- and runtime operational trustworthiness.

Unlike static operational governance approaches, PBSAI maintains:

- continuously adaptive orchestration,
- explainable remediation workflows,

- evidence-centric operational governance,
- provenance-aware authorization,
- and bounded autonomous behavior.

This enables:

- trustworthy AI-enabled enterprise operations,
- explainable response governance,
- resilient operational continuity,
- deterministic remediation control,
- and continuously adaptive enterprise AI governance management.

9. Cross-Functional AI RMF Alignment Summary

The Practitioner's Blueprint for Secure AI (PBSAI) Governance Ecosystem aligns comprehensively with the NIST Artificial Intelligence Risk Management Framework (AI RMF) through:

- deterministic governance mediation,
- evidence-centric operational accountability,
- provenance-aware execution,
- explainable orchestration,
- bounded operational autonomy,
- continuous assurance monitoring,
- and adaptive governance enforcement.

PBSAI transforms AI governance from:

- static policy documentation,
into:
- runtime operational governance.

The architecture operationalizes:

- contextual authorization,
- machine-readable governance enforcement,

- explainable operational execution,
- telemetry-aware validation,
- and resilient orchestration governance.

9.1 AI RMF Functional Alignment Summary

AI RMF Function	Primary PBSAI Domains	Primary Governance Characteristics
GOVERN	A, I, L	Governance policy, accountability, oversight, architecture governance, human authorization
MAP	B, C, F, I, K	Context awareness, identity governance, provenance, dependency mapping, data governance
MEASURE	D, G, H, L	Monitoring, telemetry analytics, evidence generation, validation, explainability
MANAGE	E, G, H, L	Remediation, resilience, orchestration, rollback, adaptive governance

9.2 PBSAI Trustworthiness Characteristic Alignment

Trustworthiness Characteristic	PBSAI Alignment Strength	Primary Supporting Capabilities
Accountability	Very Strong	Evidence generation, provenance, authorization mediation
Explainability	Strong	Structured Output Contracts, rationale preservation, telemetry analytics
Transparency	Strong	Cross-domain telemetry correlation, operational evidence reconstruction
Reliability	Very Strong	Continuous assurance monitoring, rollback, resilience orchestration
Safety	Very Strong	Bounded execution, HITL controls, kill-switch governance

Trustworthiness Characteristic	PBSAI Alignment Strength	Primary Supporting Capabilities
Security and Resilience	Very Strong	Segmentation, adaptive monitoring, recovery orchestration
Privacy Enhancement	Strong	Data governance, lawful processing validation, retention enforcement
Governability	Very Strong	Deterministic policy mediation, orchestration constraints
Traceability	Very Strong	Provenance, evidence chains, contextual governance metadata
Human Oversight	Very Strong	HITL approval enforcement, escalation governance

9.3 PBSAI Operational Governance Differentiators

PBSAI differs from traditional governance architectures by operationalizing:

- runtime governance mediation,
- bounded orchestration control,
- machine-governable execution constraints,
- structured explainability,
- provenance-aware operational validation,
- and continuously adaptive governance assurance.

Key differentiators include:

PBSAI Capability	Operational Significance
Structured Output Contracts	Deterministic operational validation of AI outputs
Context Envelope Governance	Runtime contextual governance preservation
Evidence-Centric Accountability	Explainable operational reconstruction and auditability
Provenance-Aware Execution	Attributable operational lineage and trust validation

PBSAI Capability	Operational Significance
HITL Governance	Human authorization for high-impact operational actions
Bounded Execution Semantics	Prevention of unsafe autonomous operations
Cross-Domain Telemetry Correlation	Enterprise-scale operational situational awareness
Continuous Assurance Monitoring	Adaptive trustworthiness validation
Rollback and Safe Restoration	Resilient operational recovery governance

9.4 PBSAI as an Operational Realization Architecture

PBSAI demonstrates how NIST AI RMF principles can be operationalized into:

- executable governance controls,
- explainable operational workflows,
- deterministic authorization pipelines,
- resilient orchestration frameworks,
- and evidence-centric enterprise governance systems.

The architecture therefore functions as:

- a runtime AI governance plane,
 - a deterministic operational mediation layer,
 - and an enterprise-scale trustworthy AI operationalization framework.
-

10. Conclusion

The Practitioner’s Blueprint for Secure AI (PBSAI) Governance Ecosystem provides a comprehensive operational governance architecture for AI-enabled enterprise systems.

PBSAI aligns strongly with the NIST AI Risk Management Framework through:

- governance-oriented architecture,
- explainable operational workflows,
- runtime authorization mediation,

- evidence-centric accountability,
- provenance-aware execution,
- resilience-aware orchestration,
- and continuously adaptive governance assurance.

Unlike governance approaches focused primarily on:

- policy definition,
- lifecycle documentation,
- or organizational process management,

PBSAI operationalizes trustworthy AI governance directly within runtime enterprise operations.

The architecture introduces:

- deterministic policy mediation,
- structured context governance,
- bounded operational execution,
- explainable telemetry correlation,
- human oversight enforcement,
- continuous operational validation,
- and evidence-centric orchestration governance.

PBSAI therefore serves simultaneously as:

- a practical operational realization model for trustworthy AI governance,
- a multi-agent enterprise governance architecture,
- a deterministic orchestration governance framework,
- and a structured implementation model for secure AI-enabled enterprise operations.

The architecture demonstrates how AI governance principles can evolve from:

- static governance objectives,
into:
- continuously operating governance systems capable of mediating enterprise AI behavior at runtime.

Appendix A — Relationship Semantics

Relationship Term	Meaning
equivalent	PBSAI control directly implements or operationalizes the AI RMF subcategory
strongly-supports	PBSAI control substantially contributes to satisfying the AI RMF subcategory
supports	PBSAI control partially contributes or provides supporting operational capability
enhances	PBSAI control extends the AI RMF capability with additional operational governance functionality

Appendix B — Trustworthiness Characteristic Definitions

Trustworthiness Characteristic	Definition
Accountability	Ability to attribute operational actions and governance decisions
Explainability	Ability to understand operational reasoning and decision rationale
Transparency	Visibility into operational processes, telemetry, and governance behavior
Reliability	Consistent and dependable operational behavior under expected conditions
Safety	Prevention of unsafe or harmful operational outcomes
Security and Resilience	Ability to resist, recover from, and adapt to operational threats and failures
Privacy Enhancement	Protection of sensitive and regulated data throughout operational workflows

Trustworthiness Characteristic	Definition
Governability	Ability to constrain, control, and manage operational behavior
Traceability	Ability to reconstruct operational events and governance lineage
Human Oversight	Ability for authorized humans to supervise and intervene in operational activities

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